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F70TS Time Series and Machine Learning
Coursework 2
Machine Learning

Ann Erinjeri H00418805

Introduction

Aviva insurance company is interested in knowing if their customers would be interested in taking up a new car insurance policy. This project aims to develop machine learning models to predict the chance of an existing health insurance customer buying a new car insurance policy. The data collected to create this classification system contains 12 features, including the Response variable. The dataset includes demographic data, vehicle-related data, and customer insurance information. The plan is to evaluate how well the specified models perform in predicting the response variable and to conduct experiments that may improve these models.

Task 1 — EDA

The dataset used for this study contains data collected from existing Aviva health insurance policyholders. There were no missing values in any columns. The dataset contains data on 11 features, including the response variable, which determines whether the customer opted to purchase a car insurance or not.

The distribution of the response variable is plotted in Figure 1.

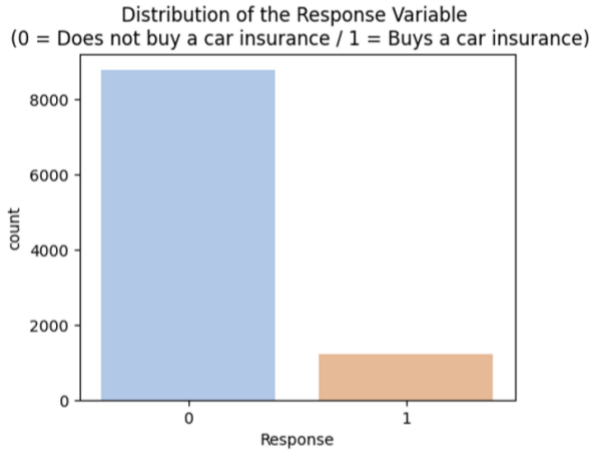

Figure 1: Distribution of the Response Variable

The plot reveals that the response variable is highly imbalanced. Approximately 87.81% of the data sample does not buy a car insurance, while only 12.19% opt to buy one. This degree of class imbalance can cause models, including k-Nearest Neighbours (kNN) and Decision Trees (DT), to bias their predictions to the majority class. This highlights the need to handle class imbalance to improve model performance and consider metrics like F1-score and recall.

The dataset features are split into two types: Numerical and Categorical.

The summary statistics of the numerical variables are highlighted in Table 1.

Variable	Mean	Std Dev	Min	25%	Median	75%	Max
Age	52.436	18.712	20	36	53	69	84
Annual Premium	28746.149	10408.322	6968.56	21360.95	26973.675	34033.2225	80000
Vintage	153.215	83.362	10	81	154	225	299

Table 1: Summary Statistics of Numerical Features (Variables)

The ages of the policyholders range between 20 and 84, with a mean of 52.436 years and a standard deviation of 18.712. The distribution of the 'age' variable is approximately uniform, indicating that the

sample represents diverse age groups evenly. The annual premium amounts paid (in currency) by the customers range from 6968.56 to 80000, with a mean of 28746.149 and a standard deviation of 10408.322. The distribution of this variable is highly skewed to the right, indicating that most policyholders pay an annual premium amount between 15000 and 40000. In contrast, a small percentage pays significantly higher amounts annually. This causes scale variation affecting distance-based models like kNN, so it must be normalized during training. The variable 'Vintage' represents the number of days the customer has been associated with the company. It ranges from 10 to 299 days, with a mean of 153.215 days and a standard deviation of 83.362. The distribution of 'Vintage' appears to be uniform, indicating that the customer groups are represented in terms of their tenure with the company. The distributions of the numerical variables are plotted in Figure 2.

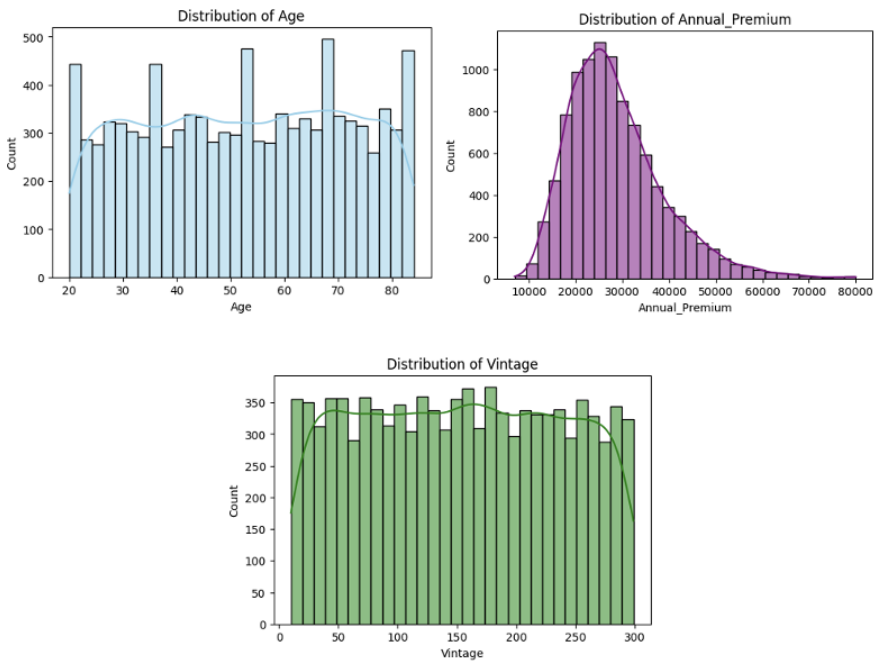

Figure 2: Distribution of Numerical Variables: Age, Annual_Premium, and Vintage

The summary statistics of the categorical, binary, and target variables are highlighted in Table 2. Approximately, 54.63% of the sample is male and 45.37% is female. A large majority (around 92.05%) possess a Driving License, while only 7.95% lack one. Around 42.54% of vehicles are 1–2 years old, 30.07% are older than 2 years, and 27.39% are less than 1 year old. About 51.01% of the sample reported no prior damage, and 48.99% reported previous damage. Approximately 55.65% of customers had no prior insurance, while 44.35% were previously insured.

The Python output carries visualizations for the individual features and how the 'Response' varies with each feature. Observing them, we realize that 'Previously Insured' has a moderately negative correlation with the 'Response' variable. This means that customers who already hold a health insurance with Aviva are unlikely to opt to buy a new car insurance. There is also a weak positive correlation between 'Vehicle_Damage' and 'Response'; customers who have experienced previous vehicle damage are slightly more likely to opt to buy a car insurance. There seems to be no statistically significant correlation between any other features and the response variable. There is no multicollinearity between any numerical features as implied by the correlation matrix in the Python output, which is a good indicator.

Variable	Category Distribution (%)	Feature Type
Response	0: 87.81%, 1: 12.19%	Target
Gender	Male: 54.63%, Female: 45.37%	Categorical
Driving License	1: 92.05%, 0: 7.95%	Binary
Vehicle Age	1–2 Years: 42.54%, > 2 Years: 30.07%, < 1 Year: 27.39%	Categorical
Vehicle Damage	No: 51.01%, Yes: 48.99%	Binary
Region Code	RG_49: 2.49%, RG_18: 2.37%, RG_22: 2.35%, RG_38: 2.27%, RG_41: 2.24%, RG_20: 2.21%, RG_01: 2.20%, RG_47: 2.16%, RG_46: 2.16%, RG_30: 2.15%, RG_11: 2.12%, RG_40: 2.11%, RG_15: 2.09%, RG_43: 2.07%, RG_23: 2.06%, RG_02: 2.05%, RG_29: 2.04%, RG_06: 1.99%, RG_36: 1.99%, RG_04: 1.99%, RG_25: 1.99%, RG_33: 1.97%, RG_42: 1.94%, RG_14: 1.94%, RG_53: 1.92%, RG_07: 1.92%, RG_24: 1.88%, RG_45: 1.88%, RG_09: 1.80%, RG_05: 1.78%, RG_17: 1.76%, RG_27: 1.74%, RG_31: 1.73%, RG_32: 1.72%, RG_10: 1.71%, RG_19: 1.71%, RG_34: 1.70%, RG_35: 1.69%, RG_16: 1.69%, RG_08: 1.68%, RG_52: 1.68%, RG_44: 1.68%, RG_39: 1.67%, RG_48: 1.64%, RG_51: 1.64%, RG_37: 1.62%, RG_26: 1.62%, RG_28: 1.59%, RG_12: 1.58%, RG_03: 1.57%, RG_21: 1.51%, RG_50: 1.49%, RG_13: 1.45%	Categorical
Previously Insured	0: 55.65%, 1: 44.35%	Binary
Policy Sales Channel	26: 43.95%, 124: 17.32%, 152: 15.96%, 156: 11.41%, 160: 11.36%	Categorical

Table 2: Summary Statistics of Categorical, Binary, and Target Features (Variables)

Task 2 — kNN from scratch

The k-Nearest Neighbours algorithm was implemented manually on the raw dataset without feature scaling. The algorithm calculates the Euclidean distances between each test sample and every training sample. The distances are sorted in ascending order, and the labels of the closest k-nearest neighbors are identified. Out of these labels, the label that is repeated the most times is selected as the predicted class (majority voting). The model was tested with values of $k \in \{1, 3, 5, 7, 9, 11\}$. Model performance was evaluated using accuracy, precision, recall, and F1-score. The model results are shown in Table 3.

k	Accuracy	Precision	Recall	F1-score
1	1.000000	1.000000	1.000000	1.000000
3	0.894875	0.697059	0.243077	0.360456
5	0.884125	0.660000	0.101538	0.176000
7	0.878875	0.562500	0.027692	0.052786
9	0.878000	0.473684	0.009231	0.018109
11	0.878125	0.500000	0.002051	0.004086

3(a)

k	Accuracy	Precision	Recall	F1-score
1	0.7730	0.102273	0.110656	0.106299
3	0.8540	0.125000	0.032787	0.051948
5	0.8690	0.125000	0.012295	0.022388
7	0.8755	0.142857	0.004098	0.007968
9	0.8765	0.000000	0.000000	0.000000
11	0.8775	0.000000	0.000000	0.000000

3(b)

Table 3: k-NN (manual implementation) training (a) and testing (b) results for $k \in \{1, 3, 5, 7, 9, 11\}$

The model exhibits moderately high accuracy for all values of k. As k increases, the accuracy increases, but the F1 and Recall drop significantly. This indicates that due to the high imbalance in the dataset, the model favors the majority class, resulting in poor detection of the minority class. Due to the absence of feature scaling, features with larger ranges (like Annual_Premium) dominate the calculation of the distances, reducing the influence of small-range and categorical variables. The model is thus unable to capture the relationships between features accurately.

Although at k=1, the model heavily overfits the training data (shown in Table 3 (a)), it is chosen as the best model. The decision is based on the F1-score and recall, given the dataset's high imbalance. The F1 score provides a better measure of performance on the minority class than accuracy. The confusion matrix for the best model (k =1) is shown in Table 4.

	Predicted 0	Predicted 1
Actual 0	1519	237
Actual 1	217	27

Table 4: Confusion Matrix for k=1 (k-NN Manual Implementation)

The model correctly predicted 1519 responses as '0'(does not buy car insurance), out of 1756 actual '0'. It only correctly predicted 27 responses as '1' (buys car insurance), out of 244 actual '1'. The model performs reasonably well on the majority class but struggles to detect the minority class because of high class imbalance. The Python output includes a classification report, which further supports these findings.

Task 3 — kNN (scikit- learn)
The kNN algorithm implemented using Scikit-learn was trained and tested on data normalized with a StandardScaler, using Euclidean distance as the default metric. The model was tested with values of $k \in \{1, 3, 5, 7, 9, 11\}$. Model performance was evaluated using accuracy, precision, recall, and F1-score. The model results are shown in Table 5.

k	Accuracy	Precision	Recall	F1-score
1	1.000000	1.000000	1.000000	1.000000
3	0.896000	0.691689	0.264615	0.382789
5	0.883750	0.647059	0.101538	0.175532
7	0.881250	0.681159	0.048205	0.090038
9	0.879875	0.750000	0.021538	0.041874
11	0.878125	0.500000	0.005128	0.010152

5(a)

k	Accuracy	Precision	Recall	F1-score
1	0.7950	0.132743	0.122951	0.127660
3	0.8455	0.164948	0.065574	0.093842
5	0.8565	0.061224	0.012295	0.020478
7	0.8705	0.105263	0.008197	0.015209
9	0.8755	0.222222	0.008197	0.015810
11	0.8765	0.000000	0.000000	0.000000

5(b)

Table 5: k-NN (Scikit-learn) training (a) and testing (b) results for $k \in \{1, 3, 5, 7, 9, 11\}$

The model shows moderately high accuracy across all values of k. It achieved higher accuracy at k=1 compared to the unscaled model. For all other k values, the unscaled model outperforms the scaled model in accuracy. However, this does not mean the unscaled model is superior. Accuracy can be misleading with imbalanced datasets, as the model often predicts the majority class. When using the kNN algorithm with Scikit-learn and normalized data, the precision, recall, and F1-score are notably higher and more stable. This indicates that the model more accurately identifies the minority class and has better generalization.

This improvement in the model can be owed to normalization, which ensures all features contribute equally to calculating distances. As a result, the model becomes more stable, reliable, and less biased.

Although at k=1, the model heavily overfits the training data (shown in Table 5 (a)), it is chosen as the best model. The decision is based on the F1-score and recall, given the dataset's high imbalance. The F1 score provides a better measure of performance on the minority class than accuracy. The confusion matrix for the best model (k =1) is shown in Table 6.

	Predicted 0	Predicted 1
Actual 0	1560	196
Actual 1	214	30

Table 6: Confusion Matrix for k=1 (k-NN Sci-kit learn)

The model correctly predicted 1560 responses as '0'(does not buy car insurance) out of 1756 actual '0' cases, outperforming the unscaled model. It only correctly predicted 30 responses as '1' (buys car insurance) out of 244 actual '1' cases, but still outperformed the unscaled model. The model performs relatively better on both classes, but still struggles to detect the minority class due to high class imbalance. The Python output includes a classification report, which further supports these findings.

Task 4 — Decision Trees

The Decision Tree algorithm implemented using Scikit-learn was trained and tested on encoded data, using 'gini' as the criterion, as it performs better when there is class imbalance (Tarun, 2024). The model was tested with values of depth ∈ {1,2,3,4,5,6,8,10,12,14,18,20,25,30,40,50,60,80}. Model performance was evaluated using accuracy, precision, recall, and F1-score. The model results are shown in Table 7.

Depth	Accuracy	Precision	Recall	F1-score
1	0.878125	0.000000	0.000000	0.000000
2	0.878125	0.000000	0.000000	0.000000
3	0.878125	0.000000	0.000000	0.000000
4	0.878500	0.800000	0.004103	0.008163
5	0.880625	0.708333	0.034872	0.066471
6	0.882500	0.830189	0.045128	0.085603
8	0.889500	0.969072	0.096410	0.175373
10	0.897250	0.952663	0.165128	0.281469
12	0.907875	0.947368	0.258462	0.406124
14	0.919375	0.966102	0.350769	0.514673
18	0.944250	0.976577	0.555897	0.708497
20	0.953875	0.977918	0.635897	0.770665
25	0.976750	0.994981	0.813333	0.895034
30	0.989750	0.997770	0.917949	0.956197
40	1.000000	1.000000	1.000000	1.000000

7(a)

Depth	Accuracy	Precision	Recall	F1-score
1	0.8780	0.000000	0.000000	0.000000
2	0.8780	0.000000	0.000000	0.000000
3	0.8780	0.000000	0.000000	0.000000
4	0.8765	0.000000	0.000000	0.000000
5	0.8740	0.277778	0.020492	0.038168
6	0.8730	0.307692	0.032787	0.059259
8	0.8665	0.171429	0.024590	0.043011
10	0.8620	0.166667	0.032787	0.054795
12	0.8540	0.166667	0.049180	0.075949
14	0.8490	0.197917	0.077869	0.111765
18	0.8275	0.165563	0.102459	0.126582
20	0.8215	0.169591	0.118852	0.139759
25	0.8015	0.144186	0.127049	0.135076
30	0.7975	0.165975	0.163934	0.164948
40	0.7900	0.156250	0.163934	0.160000

7(b)

Table 7: Decision Tree (Scikit-learn) training (a) and testing (b) results for depth ∈ {1,2,3,4,5,6,8,10,12,14,18,20,25,30,40}

The model begins to overfit as the depth increases, attaining 100% training accuracy at depth=40. Test accuracy decreases with increasing depth, while test F1 score and recall improve up to depth=30, where the minority class is finally detected. The model ranks Annual_Premium, Vintage, and Age, the numerical features, as the most important features, with these variables collectively explaining more than 54% of tree splits. The k-NN model does not explicitly identify important features, but because it is distance-based, numerical features are likely to have the greatest influence on the calculations.

Based on the F1-score, the model at depth=30 seems to be the best model, despite being severely overfitted and lacking generalization. Since the dataset is imbalanced, the minority class is only detected at deeper depths as compared to balanced datasets. From a generalization standpoint, a shallower model, such as depth=14, would be more appropriate as suggested by the performance curves. The confusion matrix for the model at depth=30 is shown in Table 8.

	Predicted 0	Predicted 1
Actual 0	1555	201
Actual 1	204	40

Table 8: Confusion Matrix for depth=30 (Decision Tree)

The model correctly predicted 1555 responses as '0' (does not buy car insurance) out of 1756 actual '0' cases, which is less than the previous model. It only correctly predicted 40 responses as '1' (buys car insurance) out of 244 actual '1' cases, but it outnumbered the k-NN model. Although there is a slight improvement in the detection of the minority class, the k-NN model generalized more reliably, indicated by the smaller train-test gap for most values of k. The Decision Tree model is more interpretable because it identifies feature importances. Decision tree model complexity increases with depth, as seen in Table 7.

Task 5 - Improvement & Experiments

<u>Effect of Feature Scaling</u>
The dataset was scaled using a StandardScaler.

a) Decision Tree
   The model remains unchanged upon scaling because it is scale-invariant.
b) K-NN
   The manually implemented k-NN model from task 2 performs identically to the scikit-learn model from task 3 after feature scaling, demonstrating improved model performance.

<u>Effect of Class Balancing</u>

The class was balanced using the Synthetic Minority Oversampling Technique (SMOTE).

a) Decision Tree
   Class balancing forces the model to learn the patterns within the synthetic, oversampled minority class, yet it still generalizes very poorly. After balancing, the model at depth=1 has the best F1 and recall, while having extremely low accuracy and precision.
b) K-NN
   The K-NN model destabilizes when SMOTE is applied. For smaller values of k, the model only predicts the majority class. At $k \geq 7$, the model only predicts the minority class, and accuracy drops. It is explained by the increased density of minority samples, influencing the nearest-neighbor relationships.

<u>Effect of Cross-Validation</u>

a) Decision Tree
   Based on the cross-validated F1 score, depth=40 is now the best model, but it highly overfits the data (100%); therefore, it is not a reliable model.
b) K-NN
   The model at k=1 is still selected as the best model after cross-validation.

<u>Effect of Feature Selection</u>

a) Decision Tree
   The decision tree at depth=30 did not benefit from feature selection. The F1 and recall dropped as compared to the full model. Feature selection removed some weak but key predictors.
   Selected Features: ['remainder__Annual_Premium', 'remainder__Vintage', 'remainder__Previously_Insured', 'cat__Policy_Sales_Channel_156', 'cat__Region_Code_RG_45', 'cat__Region_Code_RG_33', 'cat__Region_Code_RG_03']

b) K-NN

The k-NN model at k= 1 shows persistent overfitting despite feature selection. The test results are slightly worse than the original model. Feature selection does not improve the k-NN model at low k values.

Selected Features: ['num__Annual_Premium', 'cat__Policy_Sales_Channel_152', 'cat__Region_Code_RG_20', 'cat__Region_Code_RG_01', 'cat__Region_Code_RG_46', 'cat__Region_Code_RG_52', 'cat__Region_Code_RG_14', 'cat__Region_Code_RG_18', 'cat__Region_Code_RG_39', 'cat__Region_Code_RG_49', 'cat__Region_Code_RG_05', 'cat__Region_Code_RG_11', 'cat__Region_Code_RG_24', 'cat__Region_Code_RG_23', 'cat__Region_Code_RG_31']

Results for these experiments are given in the Python output. Out of all four enhancements experimented with, feature scaling proved to be the most helpful in enhancing model performance. The other experiments point to the fact that neither k-NN nor Decision Tree fits the dataset very well.

Task 6 – Reflection

This coursework shows that model behavior depends heavily on the data, especially class imbalance, and domination of large-range variables. I have learnt that accuracy alone should not be considered to assess model performance in imbalanced datasets. In each task, models that attained high accuracy levels struggled to detect the minority class. Due to the imbalance, greater importance was placed on recall and F1-score when selecting the best models.

The manual k-NN model in Task 2 showed that large-range variables affect distance calculations, leading to reduced stabilization. After scaling in Task 3, the k-NN model became more consistent at identifying minority-class samples.

In Task 4, the Decision Trees achieved higher recall and F1 at deeper depths, although they were heavily overfitted, especially at depth>30. Shallower trees had difficulty detecting the minority class, but still generalized better. This is a good practical example of the challenge in balancing bias and variance when working with imbalanced data.

The experiments in Task 5 showcased how model behaviour changes with scaling, dataset balancing, cross-validation, and feature selection. Apart from feature scaling, there was no significant model improvement caused by the experiments. This indicated that the data might not be well-suited to be fit by Decision trees or K-NN.

This coursework taught me that machine learning needs to be well thought out in its methodology, tailored to the dataset it chooses to model. The evaluation metrics need to be analyzed with much deliberation, taking into account factors such as data imbalance, large-range features, and bias-variance trade-off. Next steps would be to explore different machine learning models better suited to the dataset in predicting the response variable.

References

Tarun. (2024, July 3). Gini vs. Entropy: The ultimate split decision! DataMantra. https://datamantra.medium.com/gini-vs-entropy-the-ultimate-split-decision-51d7cf360be3